

Data Science with Python – Module 2 Exercise

Hands-on Exercise No. 2

Course: Data Science with Python

Module: Statistics & Probability

Total Marks: 10

Due Date: 23/06/2026

Instructions

Prepare your solution in MS Word.

Submit in .doc or .docx format.

Late or copied submissions will not be accepted.

Learning Outcome

After completing this exercise, you will be able to:

- Apply descriptive and inferential statistics in Python.
 - Work with probability distributions.
 - Perform hypothesis testing.
 - Visualize statistical results.
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Problem Statement

You are working as a Data Scientist for a healthcare company. The client wants statistical insights into patient recovery times. Your task is to analyze the dataset using Python and apply statistical methods.

Task Requirements

Load a dataset (for example, `patients.csv`) using Pandas.

Perform descriptive statistics on recovery times:

- Calculate mean, median, and mode.
- Compute standard deviation and variance.

Create visualizations:

- Histogram of recovery times.
- Boxplot to identify outliers.

Fit a probability distribution (such as a normal distribution) to the data using `scipy.stats`.

Perform a hypothesis test (e.g., t-test) to check whether the average recovery time is greater than 10 days.

Visualize results using appropriate charts:

- Histogram
- Boxplot
- Distribution curve

Take Screenshots of

- Python code snippets.
- Statistical outputs and test results.
- Data visualizations (graphs and distribution plots).

Submission Guidelines

Upload your .docx file with code snippets, screenshots, statistical analysis, and explanations via LMS.